TEACH YOURSELF INTERMEDIATE XII UNIT 1

Human Computer Interaction (HCI)

UX, accessibility, and Figma prototyping

What you will master

- Explain HCI fundamentals and interaction channels
- Apply accessibility (contrast, readability, inclusive navigation)
- Analyze user needs and common HCI problems
- Design and evaluate wireframes in Figma

How to use these notes

- Read one section, then close the book and explain it aloud.
- Do every *Try it* box before moving on.
- Memorize tables and truth patterns — exams love them.

1.1 HCI model

HCI designs technology for people — usable, efficient, satisfying. Channels: sight (screens), touch (tap/type), hearing (audio), voice (commands), spatial (motion/gesture).

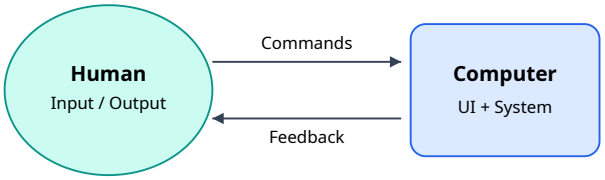


Figure 1.1 — Human sends commands; computer returns feedback through the UI.

1.2 UX vs UI

UI — visual layout, controls, typography. **UX** — entire journey (learnability, errors, satisfaction). Good UX needs user research, not only pretty screens.

1.3 Accessibility & inclusive design

- Contrast ratio for text (WCAG guidelines)
- Readable font sizes and line spacing
- Keyboard navigation and clear focus states
- Color-blind-safe palettes — don’t rely on color alone

1.4 Design process in Figma

Flow: personas → user stories → low-fi wireframes → prototype links → usability test → iterate.

Exam tip

Link each design choice to a *user need* or *accessibility principle*, not personal taste.

Teach Yourself — Practice & Check

1. Audit one school website for contrast and navigation clarity.
2. Sketch three screens for a canteen ordering app.